

Small Oscillations

(Chapter 6 Goldstein)

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1D case

$$\ddot{q} = f(q), \quad \text{EQUILIBRIUM at } q = q_0 \iff 0 = f(q_0).$$

Now we want to know what happens if we move a little bit away from q_0 :

$$q = q_0 + \eta, \quad \eta \text{ "SMALL"}.$$

$$(q_0 + \eta)'' = \ddot{\eta}, \quad f(q_0 + \eta) = \overset{0}{f(q_0)} + \eta f'(q_0) + \dots$$
$$\ddot{\eta} \simeq f'(q_0) \eta \quad \text{SMALL DEVIATION}$$

(EQUATION OF MOTION IS **LINEAR** IN SMALL DEVIATIONS)

$$\begin{array}{lll} \text{OSCILLATIONS} \iff f'(q_0) < 0, & \omega = \sqrt{-f'(q_0)}, & \eta = \eta_0 \cos(\omega t + \varphi_0), \\ \text{INSTABILITY} \iff f'(q_0) > 0, & \lambda = \sqrt{f'(q_0)}, & \eta = A e^{\lambda t} + B e^{-\lambda t}. \end{array}$$

More than one degree of freedom

(Simplify by assuming $V = V(q_1 \cdots q_n, t)$.)

$$L = T - V$$

Equilibrium at $(q_1 \cdots q_n) = (q_{01} \cdots q_{0n}) \iff$ GENERALIZED FORCES

$$Q_j = - \left. \frac{\partial V}{\partial q_j} \right|_{\vec{q}=\vec{q}_0} = 0.$$

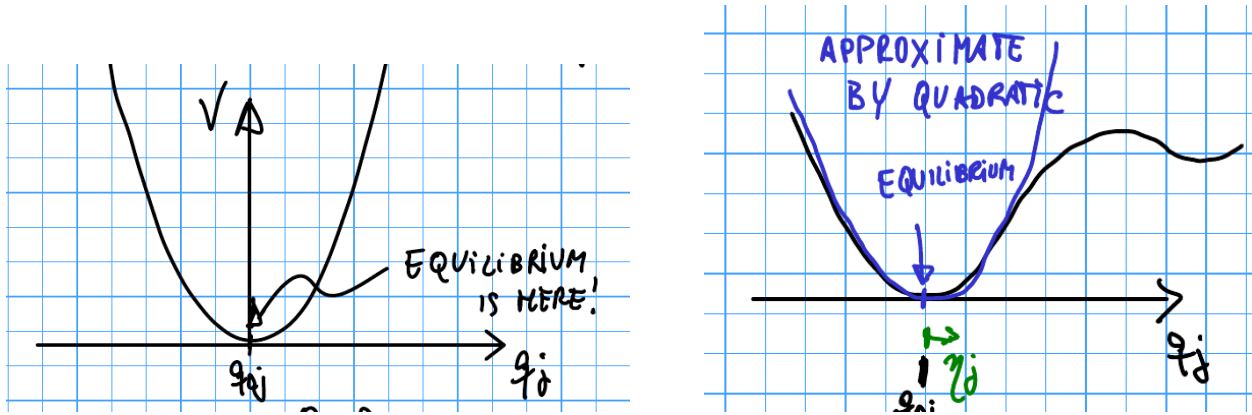


Figure 1: Left: the potential V against q_j ; the minimum at q_{0j} is marked “EQUILIBRIUM IS HERE!”. Right: the same well with the blue “APPROXIMATE BY QUADRATIC” curve laid over it, the blue arrow labelling the equilibrium point, and the green displacement η_j measured from q_{0j} — the only place in the lecture where the small-displacement variable is introduced graphically.

Taylor expand V to 2nd order in η

$$V(q_1 \cdots q_n) = \underbrace{V(q_{01} \cdots q_{0n})}_{\substack{\text{CONSTANT} \\ \Rightarrow \text{NO WORRIES}}} + \sum_{j=1}^n \underbrace{\left. \frac{\partial V}{\partial q_j} \right|_{\vec{q}_0}}_{\substack{-Q_j=0 \\ 0}} \eta_j + \frac{1}{2} \sum_{i,j=1}^n \underbrace{\left. \frac{\partial^2 V}{\partial q_i \partial q_j} \right|_{\vec{q}_0}}_{\equiv V_{ij}} \eta_i \eta_j + \mathcal{O}(\eta^3),$$

$$q_j = q_{0j} + \eta, \quad V_{ij} \text{ an } n \times n \text{ MATRIX}, \quad V_{ij} = V_{ji}.$$

$$T = \frac{1}{2} \sum_{i,j=1}^n \underbrace{\left. \frac{\partial^2 T}{\partial \dot{q}_i \partial \dot{q}_j} \right|_{\substack{\dot{\vec{q}}=0 \\ \vec{q}=\vec{q}_0}}}_{T_{ij}} \dot{\eta}_i \dot{\eta}_j, \quad T_{ij} = T_{ji}.$$

$$\boxed{\begin{aligned} T &= \sum_{p=1}^N \frac{m_p}{2} (\dot{\vec{r}}_p)^2 \\ \dot{\vec{r}}_p &= \sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t} \quad (\text{ASSUME}) \end{aligned}}$$

Lagrangian, Lagrange equations and the eigenvalue problem

$$\boxed{L = \frac{1}{2} \sum_{i,j=1}^n (T_{ij} \dot{\eta}_i \dot{\eta}_j - V_{ij} \eta_i \eta_j)}$$

$$\frac{\partial}{\partial \eta_2} \left(\frac{1}{2} [V_{11} \eta_1 \eta_1 + V_{12} \eta_1 \eta_2 + V_{21} \eta_2 \eta_1 + V_{22} \eta_2 \eta_2] \right) = \frac{1}{2} [(V_{12} + V_{21}) \eta_1 + 2V_{22} \eta_2] = V_{21} \eta_1 + V_{22} \eta_2,$$

using $V_{12} = V_{21}$.

($\frac{1}{2}$ PREFACTOR CANCELS WITH FACTOR OF 2 IN THE DERIVATIVE)

In general:

$$\frac{\partial L}{\partial \eta_i} = - \sum_{j=1}^n V_{ij} \eta_j, \quad p_i = \frac{\partial L}{\partial \dot{\eta}_i} = \sum_{j=1}^n T_{ij} \dot{\eta}_j.$$

Lagrange Eqns:

$$0 = \frac{\partial L}{\partial \eta_i} - \dot{p}_i \quad \Longrightarrow \quad \boxed{0 = \sum_{j=1}^n -V_{ij} \eta_j - \sum_{j=1}^n T_{ij} \ddot{\eta}_j} \quad i = 1, \dots, n.$$

SIMPLE CASE.

$$\begin{aligned} T &= \frac{m}{2} \dot{\eta}^2, & V &= \frac{k}{2} \eta^2, & L &= \frac{m}{2} \dot{\eta}^2 - \frac{k}{2} \eta^2, \\ \frac{\partial L}{\partial \eta} &= -k\eta, & \frac{\partial L}{\partial \dot{\eta}} &= m\dot{\eta}, \\ 0 &= \frac{\partial L}{\partial \eta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\eta}} \right) = -k\eta - m\ddot{\eta} \quad \Longrightarrow \quad \boxed{m\ddot{\eta} = -k\eta} \end{aligned}$$

Propose:

$$\eta_j = \eta_{j0} \cos(\omega t + \delta_j) \quad \Longrightarrow \quad \ddot{\eta}_j = -\omega^2 \eta_j, \quad i = 1, \dots, n.$$

$$0 = \sum_{j=1}^n (V_{ij} \eta_j - \omega^2 T_{ij} \eta_j) = \sum_{j=1}^n (V_{ij} - \omega^2 T_{ij}) \eta_{j0} \cos(\omega t + \delta_j), \quad i = 1, \dots, n.$$

BUT ALSO, since $\cos(\omega t + \delta_j) \neq 0$ for most times:

$$\boxed{0 = \sum_{j=1}^n (V_{ij} - \omega^2 T_{ij}) \eta_{j0}} \quad i = 1, \dots, n$$

EIGENVALUE EIGENVECTOR

ω : NORMAL MODE FREQUENCY η_{j0} : NORMAL MODE

For the η_{j0} to be nontrivial, we need

$$\boxed{\text{Det}[V_{ij} - \omega^2 T_{ij}] = 0}$$

Aside. More details on the Kinetic Energy part:

$$T = \sum_{p=1}^N \frac{m_p}{2} (\dot{\vec{r}}_p)^2 = \sum_{p=1}^N \frac{m_p}{2} \left(\sum_{i=1}^n \frac{\partial \vec{r}_p}{\partial q_i} \dot{q}_i + \frac{\partial \vec{r}_p}{\partial t} \right) \cdot \left(\sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t} \right)$$

$$\vec{r}_p = \vec{r}_p(q_1 \cdots q_n, t) \quad \Longrightarrow \quad \dot{\vec{r}}_p = \sum_{j=1}^n \frac{\partial \vec{r}_p}{\partial q_j} \dot{q}_j + \frac{\partial \vec{r}_p}{\partial t}$$

$$T = T_2 + T_1 + T_0$$

$$\begin{aligned}
T_2 &= \frac{1}{2} \sum_{i,j=1}^n m_{ij} \dot{q}_i \dot{q}_j, & m_{ij} &\equiv \sum_{p=1}^N m_p \frac{\partial \vec{r}_p}{\partial q_i} \cdot \frac{\partial \vec{r}_p}{\partial q_j}, \\
T_1 &= \sum_{j=1}^n \pi_j \dot{q}_j, & \pi_j &\equiv \sum_{p=1}^N m_p \frac{\partial \vec{r}_p}{\partial q_j} \cdot \frac{\partial \vec{r}_p}{\partial t}, \\
T_0 &= \sum_{p=1}^N \frac{m_p}{2} \left(\frac{\partial \vec{r}_p}{\partial t} \right)^2.
\end{aligned}$$

(m_{ij} is a function of $(q_1 \cdots q_n)$ because the derivatives $\frac{\partial \vec{r}_p}{\partial q_k}$ depend on $(q_1 \cdots q_n)$.)

Assuming $\frac{\partial \vec{r}_p}{\partial t} = \vec{0}$, and expanding for $q_j = q_{0j} + \eta_j$ with η_j small, gives

$$T = T_2 = \frac{1}{2} \sum_{i,j=1}^n \left(m_{ij}|_{\vec{q}=\vec{q}_0} + \sum_k \frac{\partial m_{ij}}{\partial q_k} \Big|_{\vec{q}=\vec{q}_0} \eta_k + \cdots \right) (q_{0i} + \eta_i)' (q_{0j} + \eta_j)'.$$

But $(q_{0i} + \eta_i)' = \dot{\eta}_i$ AND $\eta_k \dot{\eta}_i \dot{\eta}_j \propto \eta^3$ IS OF HIGHER ORDER IN SMALL DEVIATIONS.
THEREFORE, UP TO QUADRATIC ORDER IN $\eta, \dot{\eta}$:

$$T = \frac{1}{2} \sum_{i,j=1}^n T_{ij} \dot{\eta}_i \dot{\eta}_j \quad \text{WITH} \quad T_{ij} \equiv m_{ij}|_{\vec{q}=\vec{q}_0}$$

Normal modes and the modal matrix

$$\begin{aligned}
\text{Det}[V_{ij} - \omega_k^2 T_{ij}] &= 0, & k &= 1, \dots, n & (n = \# \text{ of degrees of freedom}) \\
[\mathbb{V} - \omega_k^2 \mathbb{T}] \vec{a}_k &= \vec{0}, & k &= 1, \dots, n
\end{aligned}$$

General solution is linear combination:

$$\eta_i = \sum_k C_k a_{ik} \cos(\omega_k t + \delta_k), \quad \vec{\eta} = \sum_k C_k \vec{a}_k \cos(\omega_k t + \delta_k).$$

Define $(\mathbb{A})_{ij} \equiv a_{ij}$, $\mathbb{A} \equiv (\vec{a}_1 \ \vec{a}_2 \ \cdots \ \vec{a}_n)$.

Property:

$$\mathbb{1} = \mathbb{A}^\dagger \mathbb{T} \mathbb{A} \quad \implies \quad \mathbb{A}^{-1} = \mathbb{A}^\top \mathbb{T}, \quad \mathbb{A}^\dagger \mathbb{V} \mathbb{A} = \begin{pmatrix} \omega_1^2 & 0 & \cdots & 0 \\ 0 & \ddots & & \vdots \\ \vdots & & \ddots & 0 \\ 0 & \cdots & 0 & \omega_n^2 \end{pmatrix}.$$

$\mathbb{A}$ is the matrix of eigenvectors / normal modes.

Normal modes example

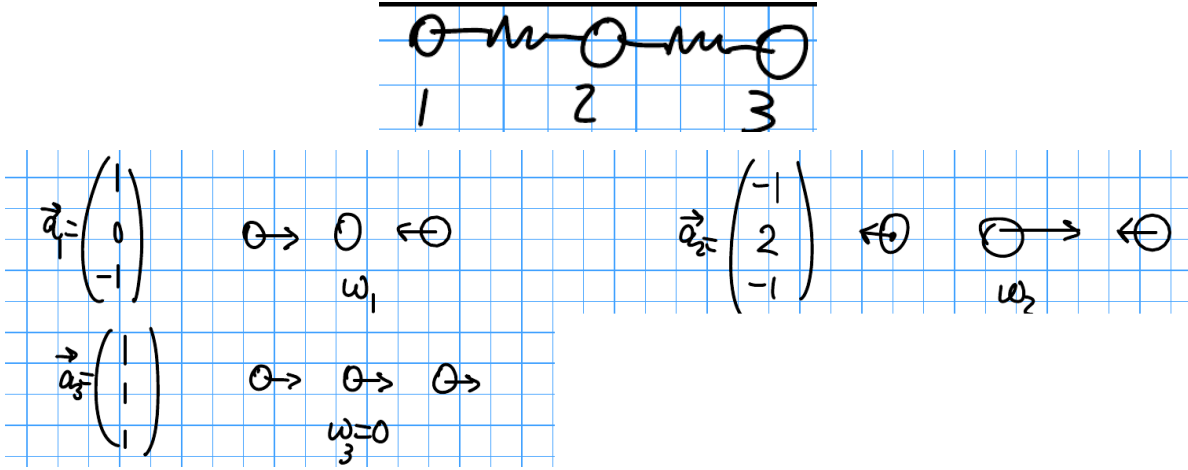


Figure 2: Three masses 1, 2, 3 coupled by two springs, and the three normal modes: $\vec{a}_1 = (1, 0, -1)^\top$ with frequency ω_1 (outer masses move oppositely, centre mass at rest); $\vec{a}_2 = (-1, 2, -1)^\top$ with ω_2 ; and the uniform translation $\vec{a}_3 = (1, 1, 1)^\top$ with $\omega_3 = 0$.

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = C_1 \vec{a}_1 \cos(\omega_1 t + \delta_1) + C_2 \vec{a}_2 \cos(\omega_2 t + \delta_2) + C_3 \vec{a}_3 \cos(\omega_3 t + \delta_3)$$

Normal coordinate ζ

$$\vec{\eta} = \mathbb{A} \vec{\zeta}, \quad \eta_i = \sum_k \zeta_k a_{ik}, \quad \zeta_k = C_k \cos(\omega_k t + \delta_k), \quad \vec{\zeta} = \mathbb{A}^{-1} \vec{\eta} = \mathbb{A}^\top \mathbb{T} \vec{\eta}.$$

$$L = \frac{1}{2} \sum_{i,j} (\dot{\eta}_i T_{ij} \dot{\eta}_j - \eta_i V_{ij} \eta_j) = \frac{1}{2} [(\dot{\vec{\eta}})^\top \mathbb{T} \dot{\vec{\eta}} - \vec{\eta}^\top \mathbb{V} \vec{\eta}] = \frac{1}{2} [\dot{\vec{\zeta}}^\top \mathbb{A}^\top \mathbb{T} \mathbb{A} \dot{\vec{\zeta}} - \vec{\zeta}^\top \mathbb{A}^\top \mathbb{V} \mathbb{A} \vec{\zeta}]$$

$$L = \frac{1}{2} \left[\dot{\vec{\zeta}}^\top \dot{\vec{\zeta}} - \vec{\zeta}^\top \begin{pmatrix} \omega_1^2 & & 0 \\ & \ddots & \\ 0 & & \omega_n^2 \end{pmatrix} \vec{\zeta} \right] \implies \boxed{L = \frac{1}{2} \sum_{k=1}^n (\dot{\zeta}_k^2 - \omega_k^2 \zeta_k^2)}$$

Example: Oscillations of a double pendulum

Transformation equations

$$\begin{aligned} \vec{r}_1 &= l(\sin \phi_1 \hat{x} - \cos \phi_1 \hat{z}) = l \hat{r}_1, \\ \vec{r}_2 &= l(\sin \phi_1 \hat{x} - \cos \phi_1 \hat{z}) + l(\sin \phi_2 \hat{x} - \cos \phi_2 \hat{z}) = l(\hat{r}_1 + \hat{r}_{21}), \\ \hat{r}_1 &= \sin \phi_1 \hat{x} - \cos \phi_1 \hat{z}, \quad \hat{r}_{21} = \sin \phi_2 \hat{x} - \cos \phi_2 \hat{z}. \end{aligned}$$

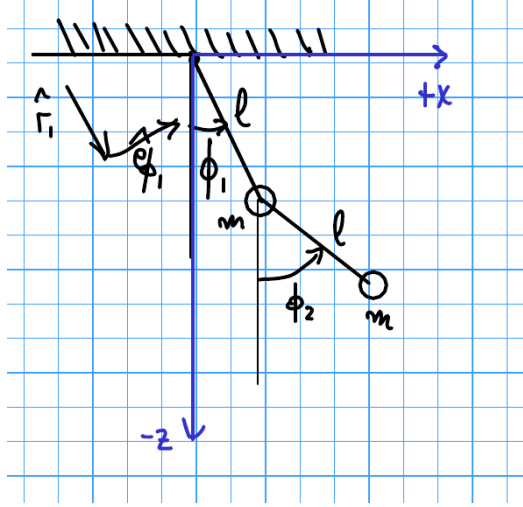


Figure 3: Two equal masses m on two rods of equal length l hung from a fixed ceiling, with angles ϕ_1, ϕ_2 measured from the downward vertical, and the unit vectors $\hat{r}_1, \hat{e}_{\phi_1}$. The blue axes fix the sign convention: $+x$ to the right and $-z$ downward, which is what makes $V = -mgl(2 \cos \phi_1 + \cos \phi_2)$.

Velocities

$$\begin{aligned}\dot{\hat{r}}_1 &= \dot{\phi}_1 \frac{\partial \hat{r}_1}{\partial \phi_1}, & \frac{\partial \hat{r}_1}{\partial \phi_1} &= \hat{e}_{\phi_1} = \cos \phi_1 \hat{x} + \sin \phi_1 \hat{z}, \\ \dot{\hat{r}}_{21} &= \dot{\phi}_2 \frac{\partial \hat{r}_{21}}{\partial \phi_2}, & \frac{\partial \hat{r}_{21}}{\partial \phi_2} &= \hat{e}_{\phi_2} = \cos \phi_2 \hat{x} + \sin \phi_2 \hat{z},\end{aligned}$$

$$\begin{aligned}\dot{\vec{r}}_1 &= l \dot{\hat{r}}_1 = l \dot{\phi}_1 (\cos \phi_1 \hat{x} + \sin \phi_1 \hat{z}), \\ \dot{\vec{r}}_2 &= l (\dot{\hat{r}}_1 + \dot{\hat{r}}_{21}) = l \dot{\phi}_1 (\cos \phi_1 \hat{x} + \sin \phi_1 \hat{z}) + l \dot{\phi}_2 (\cos \phi_2 \hat{x} + \sin \phi_2 \hat{z}).\end{aligned}$$

$$\begin{aligned}T &= \frac{m}{2} [(\dot{\vec{r}}_1)^2 + (\dot{\vec{r}}_2)^2] = \frac{m}{2} [l^2 \dot{\phi}_1^2 + l^2 (\dot{\phi}_1^2 + \dot{\phi}_2^2 + 2 \dot{\phi}_1 \dot{\phi}_2 \hat{e}_{\phi_1} \cdot \hat{e}_{\phi_2})], \\ \hat{e}_{\phi_1} \cdot \hat{e}_{\phi_2} &= \cos \phi_1 \cos \phi_2 + \sin \phi_1 \sin \phi_2 = \cos(\phi_2 - \phi_1), \\ T &= \frac{ml^2}{2} [2 \dot{\phi}_1^2 + \dot{\phi}_2^2 + 2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_2 - \phi_1)], \\ V &= mgz_1 + mgz_2 = -mgl(2 \cos \phi_1 + \cos \phi_2), \\ L &= T - V = \frac{ml^2}{2} [2 \dot{\phi}_1^2 + \dot{\phi}_2^2 + 2 \dot{\phi}_1 \dot{\phi}_2 \cos(\phi_2 - \phi_1)] + mgl(2 \cos \phi_1 + \cos \phi_2).\end{aligned}$$

We know $\phi_1 = \phi_2 = 0$ is stable equilibrium position, but let's check:¹

$$\frac{\partial V}{\partial \phi_1} = mgl \sin \phi_1 = 0 \quad \text{at } \phi_1 = 0 \quad \checkmark \qquad \frac{\partial V}{\partial \phi_2} = mgl \sin \phi_2 = 0 \quad \text{at } \phi_2 = 0 \quad \checkmark$$

And V is absolute minimum at $\phi_1 = \phi_2 = 0 \Rightarrow$ Equil is STABLE.

¹ *Transcriber's note.* The first derivative below is as written. For the V above, $\partial V / \partial \phi_1 = 2mgl \sin \phi_1$; the factor 2 is absent in the manuscript. The conclusion — a stationary point at $\phi_1 = 0$ — is unaffected.

We thus expand T and V to quadratic order in $\eta_i = \phi_i - \phi_i^0 = \phi_i$, since $\phi_i^0 = 0$:

$$\dot{\phi}_1 \dot{\phi}_2 \cos(\phi_2 - \phi_1) = \dot{\eta}_1 \dot{\eta}_2 \cos(\eta_2 - \eta_1) \simeq \dot{\eta}_1 \dot{\eta}_2 \left[1 - \frac{(\eta_2 - \eta_1)^2}{2} + \dots \right] \simeq \dot{\eta}_1 \dot{\eta}_2 + \mathcal{O}(\dot{\eta}^2 \eta^2),$$

small η_i

$$\cos \phi_i = \cos \eta_i = 1 - \frac{\eta_i^2}{2} + \frac{\eta_i^4}{4!} - \dots \simeq 1 - \frac{\eta_i^2}{2}.$$

Thus, to quadratic order in η and $\dot{\eta}$ we get

$$V = -mgl \left[3 - \frac{2\eta_1^2}{2} - \frac{\eta_2^2}{2} \right] \implies \mathbb{V} = mgl \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix},$$

$$T = \frac{ml^2}{2} [2\dot{\eta}_1^2 + \dot{\eta}_2^2 + 2\dot{\eta}_1 \dot{\eta}_2] \implies \mathbb{T} = ml^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix},$$

consistent with $T = \frac{1}{2} (\dot{\eta}_1 \dots \dot{\eta}_n) \mathbb{T} (\dot{\eta}_1 \dots \dot{\eta}_n)^\top$.

CHECK.

$$\begin{aligned} (\eta_1 \quad \eta_2) \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} &= (\eta_1 \quad \eta_2) \begin{pmatrix} 2\eta_1 \\ \eta_2 \end{pmatrix} = 2\eta_1^2 + \eta_2^2 \quad \checkmark \\ (\dot{\eta}_1 \quad \dot{\eta}_2) \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \dot{\eta}_1 \\ \dot{\eta}_2 \end{pmatrix} &= (\dot{\eta}_1 \quad \dot{\eta}_2) \begin{pmatrix} 2\dot{\eta}_1 + \dot{\eta}_2 \\ \dot{\eta}_1 + \dot{\eta}_2 \end{pmatrix} = 2\dot{\eta}_1^2 + 2\dot{\eta}_1 \dot{\eta}_2 + \dot{\eta}_2^2 \quad \checkmark \end{aligned}$$

Eigenfrequencies and eigenvectors

To simplify the algebra, we write

$$mgl = ml^2 \left(\frac{g}{l} \right), \quad \omega_0^2 \equiv \frac{g}{l}.$$

Eigenvalue equation:

$$\begin{aligned} 0 &= \text{Det}[\mathbb{V} - \omega^2 \mathbb{T}] = \text{Det} \left(ml^2 \begin{bmatrix} 2\frac{g}{l} - 2\omega^2 & -\omega^2 \\ -\omega^2 & \frac{g}{l} - \omega^2 \end{bmatrix} \right) \\ \iff 0 &= (2\omega_0^2 - 2\omega^2)(\omega_0^2 - \omega^2) - \omega^4 \\ &= 2\omega^4 - 4\omega^2\omega_0^2 + 2\omega_0^4 - \omega^4 = \omega^4 - 4\omega_0^2\omega^2 + 2\omega_0^4 \\ \iff 0 &= \underbrace{(\omega^2 - 2\omega_0^2)^2}_{\text{COMPLETE SQUARES}} - \underbrace{(4\omega_0^4 - 2\omega_0^4)}_{2\omega_0^4} \\ \iff \omega^2 &= 2\omega_0^2 \pm \sqrt{2}\omega_0^2 = (2 \pm \sqrt{2})\omega_0^2 \end{aligned}$$

$$\iff \boxed{\omega = \begin{cases} \omega_+ = \sqrt{2 + \sqrt{2}} \omega_0 \\ \omega_- = \sqrt{2 - \sqrt{2}} \omega_0 \end{cases}} \quad \text{Eigenfrequencies}$$

Eigenvectors:

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2\omega_0^2 - 2\omega^2 & -\omega^2 \\ -\omega^2 & \omega_0^2 - \omega^2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \iff \begin{cases} 0 = 2(\omega_0^2 - \omega^2)a_1 - \omega^2 a_2 \\ 0 = -\omega^2 a_1 + (\omega_0^2 - \omega^2)a_2 \end{cases}$$

$$\iff \begin{cases} a_2 = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) a_1 \\ a_1 = \left(\frac{\omega_0^2}{\omega^2} - 1\right) a_2 \end{cases} \iff \begin{cases} \frac{a_2}{a_1} = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) \\ \frac{a_2}{a_1} = \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1\right)} \end{cases}$$

SIDE NOTE: the 2 equations are equivalent:

$$\begin{aligned} 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) &= \frac{1}{\left(\frac{\omega_0^2}{\omega^2} - 1\right)} \iff 1 = 2\left(\frac{\omega_0^2}{\omega^2} - 1\right)\left(\frac{\omega_0^2}{\omega^2} - 1\right) \\ &\iff \omega^4 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) \\ &\iff 0 = 2(\omega_0^2 - \omega^2)(\omega_0^2 - \omega^2) - \omega^4 \iff 0 = \text{Det}(\mathbb{V} - \omega^2 \mathbb{T}) \checkmark \end{aligned}$$

So we can take either equation and use it, together with normalization, to get $\vec{a}$:

$$\text{Propose } \vec{a} = \mathcal{N} \begin{pmatrix} 1 \\ 2\left(\frac{\omega_0^2}{\omega^2} - 1\right) \end{pmatrix}, \quad \mathcal{N} : \text{normalization constant}, \quad x = \frac{\omega_0^2}{\omega^2}.$$

$$\begin{aligned} 1 &= [\vec{a}]^T \mathbb{T} [\vec{a}] = \mathcal{N}^2 \begin{pmatrix} 1 & 2(x-1) \end{pmatrix} ml^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2(x-1) \end{pmatrix} = \mathcal{N}^2 ml^2 \begin{pmatrix} 1 & 2(x-1) \end{pmatrix} \begin{pmatrix} 2x \\ 2x-1 \end{pmatrix} \\ &= \mathcal{N}^2 ml^2 [2x + 2(x-1)(2x-1)] = 2\mathcal{N}^2 ml^2 [2x^2 - 3x + x + 1] \\ &= 2\mathcal{N}^2 ml^2 [2x^2 - 2x + 1] = 2\mathcal{N}^2 ml^2 [2(x-1)^2 + 4x - 2 + 1] \\ &= 2\mathcal{N}^2 ml^2 \left\{ \underbrace{[2(x-1)^2 - 1]}_0 + 2x \right\} = 4\mathcal{N}^2 ml^2 \frac{\omega_0^2}{\omega^2} \\ &\implies \boxed{\mathcal{N}_{\pm} = \frac{1}{\sqrt{4m}} \frac{\omega_{\pm}}{l \omega_0} = \left(\frac{2 \pm \sqrt{2}}{4ml^2} \right)^{1/2}} \end{aligned}$$

Supporting algebra:²

$$\begin{aligned} (x-1)^2 &= x^2 - 2x + 1, & x^2 &= (x-1)^2 + 2x - 1, \\ 2x^2 - 2x + 1 &= 2[(x-1)^2 + 2x - 1] - 2x + 1 = [2(x-1)^2 - 1] + 2x, \\ \omega^4 [2(x-1)^2 - 1] &= [2(x\omega^2 - \omega^2)^2 - \omega^4] = [2(\omega_0^2 - \omega^2)^2 - \omega^4] = 0. \end{aligned}$$

$$\frac{\omega_0^2}{\omega_{\pm}^2} = \frac{1}{2 \pm \sqrt{2}} = \frac{2 \mp \sqrt{2}}{4 - 2} = 1 \mp \frac{1}{\sqrt{2}}, \quad 2 \left[\frac{\omega_0^2}{\omega_{\pm}^2} - 1 \right] = \mp \sqrt{2},$$

$$\boxed{\vec{a}_{\pm} = \left(\frac{2 \pm \sqrt{2}}{4ml^2} \right)^{1/2} \begin{pmatrix} 1 \\ \mp \sqrt{2} \end{pmatrix}} \iff \boxed{\omega_{\pm} = \left[(2 \pm \sqrt{2}) \frac{g}{l} \right]^{1/2}}$$

² *Transcriber's note.* In the third line above, the term $-2x$ is absent from the bracket $[2(x-1)^2 + 4x - 2 + 1]$; transcribed as written. The lecturer's own side computation immediately below carries it: $2x^2 - 2x + 1 = 2[(x-1)^2 + 2x - 1] - 2x + 1 = [2(x-1)^2 - 1] + 2x$, which is what the next equality uses.

Modal matrix:

$$\mathbb{A} = (\vec{a}_+ \quad \vec{a}_-) = (4ml^2)^{-1/2} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$a \equiv (2 + \sqrt{2})^{1/2} = -c/\sqrt{2},$$

$$c = -\sqrt{2}a = -\sqrt{2}(2 + \sqrt{2})^{1/2},$$

$$b \equiv (2 - \sqrt{2})^{1/2} = d/\sqrt{2},$$

$$d = +\sqrt{2}b = +\sqrt{2}(2 - \sqrt{2})^{1/2}.$$

Graphical representation

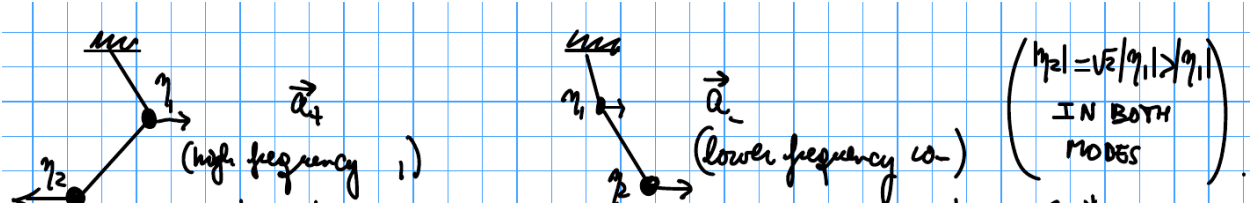


Figure 4: The two normal modes of the double pendulum. Left: $\vec{a}_+$, the high-frequency mode ω_+ , in which the two bobs move in opposite senses. Right: $\vec{a}_-$, the lower-frequency mode ω_- , in which they move together. In both modes $|\eta_2| = \sqrt{2}|\eta_1| > |\eta_1|$.

Normal coordinates

$$\vec{\zeta} = \mathbb{A}^T \mathbb{T} \vec{\eta} = \mathbb{A}^T ml^2 \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix}$$

$$\vec{\zeta} = (4ml^2)^{-1/2} \begin{pmatrix} a & c \\ b & d \end{pmatrix} ml^2 \begin{pmatrix} 2\eta_1 + \eta_2 \\ \eta_1 + \eta_2 \end{pmatrix} = \left(\frac{ml^2}{4} \right)^{1/2} \begin{pmatrix} (2a + c)\eta_1 + (a + c)\eta_2 \\ (2b + d)\eta_1 + (b + d)\eta_2 \end{pmatrix}$$

with

$$2a + c = (2 - \sqrt{2})a = \sqrt{2}(\sqrt{2} - 1)^{1/2}(\sqrt{2} - 1)^{1/2}(2 + \sqrt{2})^{1/2} = \sqrt{2} 2^{1/4}(\sqrt{2} - 1)^{1/2},$$

$$a + c = (1 - \sqrt{2})a = -(\sqrt{2} - 1)^{1/2}(\sqrt{2} - 1)^{1/2}(2 + \sqrt{2})^{1/2} = -2^{1/4}(\sqrt{2} - 1)^{1/2},$$

$$2b + d = (2 + \sqrt{2})b = \sqrt{2}(\sqrt{2} + 1)^{1/2}(\sqrt{2} + 1)^{1/2}(2 - \sqrt{2})^{1/2} = \sqrt{2} 2^{1/4}(\sqrt{2} + 1)^{1/2},$$

$$b + d = (1 + \sqrt{2})b = (\sqrt{2} + 1)^{1/2}(\sqrt{2} + 1)^{1/2}(2 - \sqrt{2})^{1/2} = 2^{1/4}(\sqrt{2} + 1)^{1/2},$$

where

$$(\sqrt{2} \mp 1)^{1/2}(2 \pm \sqrt{2})^{1/2} = 2^{1/4}(\sqrt{2} \mp 1)^{1/2}(\sqrt{2} \pm 1)^{1/2} = 2^{1/4},$$

$$\left(\frac{ml^2}{4} \right)^{1/2} = m^{1/2} \frac{l}{2}, \quad \frac{1}{2} 2^{1/4}(\sqrt{2} \mp 1)^{1/2} = \left(\frac{\sqrt{2} \mp 1}{2\sqrt{2}} \right)^{1/2}.$$

$$\begin{aligned}\zeta_+(t) &= m^{1/2}l \left(\frac{\sqrt{2}-1}{2\sqrt{2}} \right)^{1/2} [\sqrt{2}\eta_1(t) - \eta_2(t)] \\ \zeta_-(t) &= m^{1/2}l \left(\frac{\sqrt{2}+1}{2\sqrt{2}} \right)^{1/2} [\sqrt{2}\eta_1(t) + \eta_2(t)]\end{aligned}$$

$$\begin{aligned}\dot{\zeta}_+(t) &= m^{1/2}l \left(\frac{\sqrt{2}-1}{2\sqrt{2}} \right)^{1/2} [\sqrt{2}\dot{\eta}_1(t) - \dot{\eta}_2(t)] \\ \dot{\zeta}_-(t) &= m^{1/2}l \left(\frac{\sqrt{2}+1}{2\sqrt{2}} \right)^{1/2} [\sqrt{2}\dot{\eta}_1(t) + \dot{\eta}_2(t)]\end{aligned}$$

But we know also that:

$$\begin{aligned}\zeta_+(t) &= f_+ \cos(\omega_+ t + \delta_+) & \dot{\zeta}_+(t) &= -f_+ \omega_+ \sin(\omega_+ t + \delta_+) \\ \zeta_-(t) &= f_- \cos(\omega_- t + \delta_-) & \dot{\zeta}_-(t) &= -f_- \omega_- \sin(\omega_- t + \delta_-)\end{aligned}$$

Lagrangian in terms of the normal coordinates

$$\begin{aligned}L &= \frac{1}{2} \sum_{\alpha} (\dot{\zeta}_{\alpha}^2 - \omega_{\alpha}^2 \zeta_{\alpha}^2) = \frac{1}{2} [(\dot{\zeta}_+^2 - \omega_+^2 \zeta_+^2) + (\dot{\zeta}_-^2 - \omega_-^2 \zeta_-^2)] \\ L &= \frac{1}{2} \left[\left(\dot{\zeta}_+^2 - \left[(2 + \sqrt{2}) \frac{g}{l} \right] \zeta_+^2 \right) + \left(\dot{\zeta}_-^2 - \left[(2 - \sqrt{2}) \frac{g}{l} \right] \zeta_-^2 \right) \right]\end{aligned}$$

Solution for a given initial condition

Assume initial condition is that at $t = 0$:

$$\phi_1 = \phi_2 = 0, \quad \dot{\phi}_1 = -\dot{\phi}_2 = \nu > 0.$$

Get $\phi_1(t)$, $\phi_2(t)$. Then

$$\begin{aligned}f_+ \cos(\delta_+) = \zeta_+(0) = 0 &\implies \delta_+ = \pm \frac{\pi}{2}, \pm \frac{3}{2}\pi, \dots \\ f_- \cos(\delta_-) = \zeta_-(0) = 0 &\implies \delta_- = \pm \frac{\pi}{2}, \pm \frac{3}{2}\pi, \dots\end{aligned}$$

$$\begin{aligned}-f_+ \omega_+ \sin(\delta_+) = \dot{\zeta}_+(0) &= m^{1/2}l \left(\frac{\sqrt{2}-1}{2\sqrt{2}} \right)^{1/2} [\sqrt{2}\nu - (-\nu)] = \sqrt{2+\sqrt{2}} \frac{m^{1/2}l\nu}{2}, \\ -f_- \omega_- \sin(\delta_-) = \dot{\zeta}_-(0) &= m^{1/2}l \left(\frac{\sqrt{2}+1}{2\sqrt{2}} \right)^{1/2} [\sqrt{2}\nu + (-\nu)] = \sqrt{2-\sqrt{2}} \frac{m^{1/2}l\nu}{2},\end{aligned}$$

using

$$m^{1/2}l \left(\frac{\sqrt{2} \mp 1}{2\sqrt{2}} \right)^{1/2} (\sqrt{2} \pm 1)\nu = \left(\frac{\sqrt{2} \pm 1}{2\sqrt{2}} \right)^{1/2} m^{1/2}l \nu = \sqrt{2 \pm \sqrt{2}} \frac{m^{1/2}l \nu}{2}.$$

Choose $\delta_+ = \delta_- = -\frac{\pi}{2}$:

$$\Rightarrow f_{\pm} \sqrt{2 \pm \sqrt{2}} \sqrt{\frac{g}{l}} = \sqrt{2 \pm \sqrt{2}} \frac{m^{1/2}l \nu}{2} \Rightarrow \boxed{f_{\pm} = \frac{\nu}{2} \left[\frac{ml^3}{g} \right]^{1/2}}$$

$$\cos\left(\omega t - \frac{\pi}{2}\right) = \cos \omega t \overset{0}{\cos} \frac{\pi}{2} + \sin \omega t \underset{1}{\sin} \frac{\pi}{2} = \sin \omega t$$

$$\zeta_{\pm}(t) = f_{\pm} \cos\left(\omega_{\pm} t - \frac{\pi}{2}\right) = f_{\pm} \sin(\omega_{\pm} t)$$

$$\boxed{\zeta_{\pm}(t) = \frac{\nu}{2} \left[\frac{ml^3}{g} \right]^{1/2} \sin\left(\left[(2 \pm \sqrt{2}) \frac{g}{l}\right]^{1/2} t\right)}$$

with

$$a \equiv (2 + \sqrt{2})^{1/2} = 2^{1/4}(\sqrt{2} + 1)^{1/2}, \quad c = -\sqrt{2}a, \quad f_+ a = \frac{\nu}{2} \left[\frac{ml^3}{g} \right]^{1/2} (2 + \sqrt{2})^{1/2},$$

$$b \equiv (2 - \sqrt{2})^{1/2} = 2^{1/4}(\sqrt{2} - 1)^{1/2}, \quad d = +\sqrt{2}b, \quad f_- b = \frac{\nu}{2} \left[\frac{ml^3}{g} \right]^{1/2} (2 - \sqrt{2})^{1/2},$$

$$(4ml^2)^{-1/2} \frac{\nu}{2} \left(\frac{ml^3}{g} \right)^{1/2} = \frac{\nu}{4\sqrt{\frac{g}{l}}}.$$

In terms of the original coordinates

$$\vec{\eta}(t) = \mathbb{A} \vec{\zeta}(t) = (4ml^2)^{-1/2} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \zeta_+(t) \\ \zeta_-(t) \end{pmatrix} = (4ml^2)^{-1/2} \begin{pmatrix} a \zeta_+(t) + b \zeta_-(t) \\ c \zeta_+(t) + d \zeta_-(t) \end{pmatrix}$$

$$\boxed{\begin{aligned} \phi_1(t) = \eta_1(t) &= \frac{\nu}{4\sqrt{\frac{g}{l}}} \left\{ (2 + \sqrt{2})^{1/2} \sin\left(\left[(2 + \sqrt{2}) \frac{g}{l}\right]^{1/2} t\right) \right. \\ &\quad \left. + (2 - \sqrt{2})^{1/2} \sin\left(\left[(2 - \sqrt{2}) \frac{g}{l}\right]^{1/2} t\right) \right\} \\ \phi_2(t) = \eta_2(t) &= \frac{\sqrt{2}\nu}{4\sqrt{\frac{g}{l}}} \left\{ -(2 + \sqrt{2})^{1/2} \sin\left(\left[(2 + \sqrt{2}) \frac{g}{l}\right]^{1/2} t\right) \right. \\ &\quad \left. + (2 - \sqrt{2})^{1/2} \sin\left(\left[(2 - \sqrt{2}) \frac{g}{l}\right]^{1/2} t\right) \right\} \end{aligned}}$$

[typed annotation on the manuscript page — not in the lecturer's hand]

Check initial condition: At $t=0$, all $\sin(\dots)$ terms are zero, so we have:

$$\phi_1 = \phi_2 = 0 \quad \checkmark$$

$$\begin{aligned} \dot{\phi}_1(t) = \dot{\eta}_1(t) &= \frac{\nu}{4} \left\{ (2 + \sqrt{2}) \cos \left(\left[(2 + \sqrt{2}) \frac{g}{l} \right]^{1/2} t \right) \right. \\ &\quad \left. + (2 - \sqrt{2}) \cos \left(\left[(2 - \sqrt{2}) \frac{g}{l} \right]^{1/2} t \right) \right\} \\ \dot{\phi}_2(t) = \dot{\eta}_2(t) &= \frac{\sqrt{2}\nu}{4} \left\{ -(2 + \sqrt{2}) \cos \left(\left[(2 + \sqrt{2}) \frac{g}{l} \right]^{1/2} t \right) \right. \\ &\quad \left. + (2 - \sqrt{2}) \cos \left(\left[(2 - \sqrt{2}) \frac{g}{l} \right]^{1/2} t \right) \right\} \end{aligned}$$

[typed annotation on the manuscript page — not in the lecturer's hand]

Check initial condition: At $t=0$, all $\cos(\dots)$ terms are 1, so we have:

$$\dot{\phi}_1 = \frac{\nu}{4} \left\{ (2 + \sqrt{2}) + (2 - \sqrt{2}) \right\} = \nu \quad \checkmark$$

$$\dot{\phi}_2 = \frac{\sqrt{2}\nu}{4} \left\{ -(2 + \sqrt{2}) + (2 - \sqrt{2}) \right\} = -\nu \quad \checkmark$$

Comparison with quantum mechanics

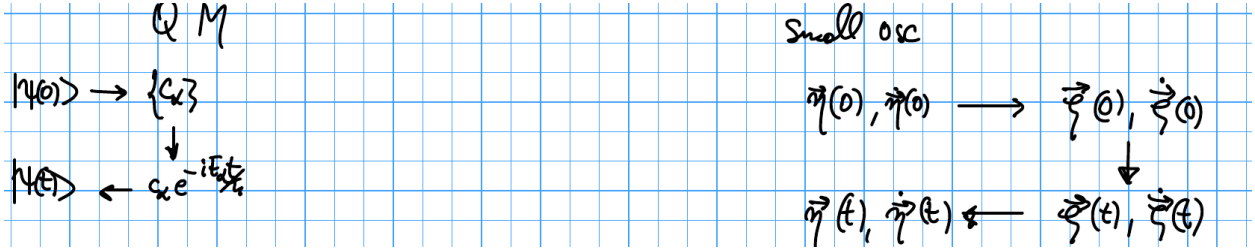


Figure 5: The two schemes side by side. Left (QM): the initial state $|\psi(0)\rangle$ is resolved into the coefficients $\{c_\alpha\}$, each is propagated by $c_\alpha e^{-iE_\alpha t/\hbar}$, and the result is reassembled into $|\psi(t)\rangle$. Right (small osc): $\vec{\eta}(0), \dot{\vec{\eta}}(0)$ are mapped to $\vec{\zeta}(0), \dot{\vec{\zeta}}(0)$, propagated as $\vec{\zeta}(t), \dot{\vec{\zeta}}(t)$, and mapped back.

QM

$$|\psi(0)\rangle = \sum_{\alpha} c_{\alpha} |\varphi_{\alpha}\rangle, \quad c_{\alpha} = \langle \varphi_{\alpha} | \psi(0) \rangle$$

$$|\psi(t)\rangle = \sum_{\alpha} c_{\alpha} e^{-iE_{\alpha} t/\hbar} |\varphi_{\alpha}\rangle$$

Small osc

$$\vec{\eta}(t) = \mathbb{A} \vec{\zeta}(t)$$

$$\vec{\zeta}(t) = \mathbb{A}^T \mathbb{T} \vec{\eta}(t)$$